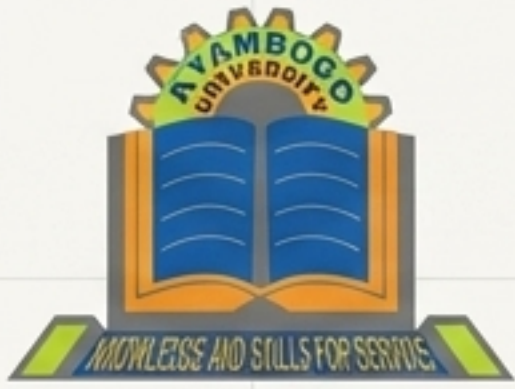




GSM Signal Denoising & Modulation Classification

Improving Connectivity in Underserved Areas
Using Machine Learning

Target: Rural Uganda Sub-counties

Pipeline: Hybrid DAE-AMC

Researchers: Ssemuju Sharif Abdukarim &

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Supervisor: Dr. Dickson Mugerwa

Department: Electrical and Electronics

Engineering, Kyambogo University

Exponential Demand

33.2 Million

Network Subscriptions in 2022

421.5 Million GBs

Mobile data traffic exchanged, doubling since 2020

The Rural Deficit

62%

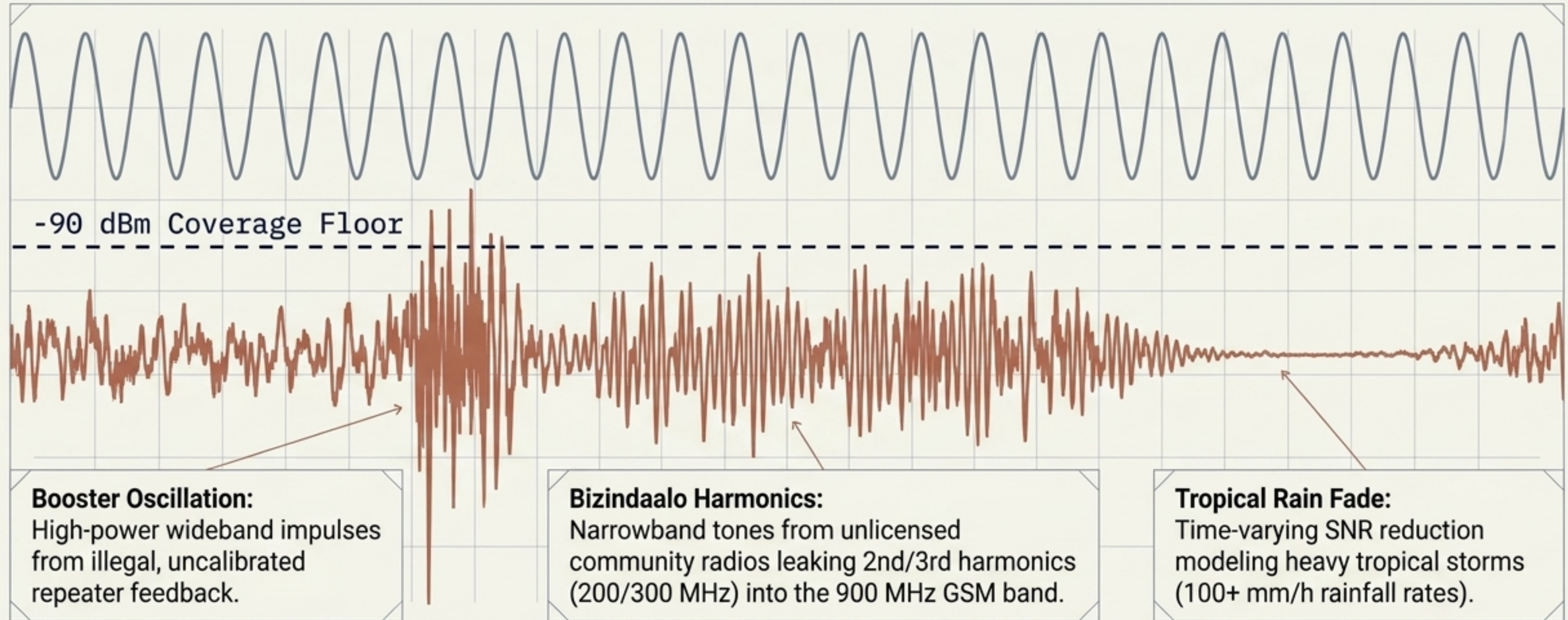
Of Ugandans lack mobile internet access
(approx. 30 million people)

31%

Population with 5G/4G coverage

Most rural users still rely on basic phones for calls and vital Mobile Money, trapped in **low SNR, high-interference environments** where standard network expansion is **failing to keep pace**.

The Anatomy of Rural Interference



In environments below -90 dBm, the noise becomes louder than the message. Standard demodulation tools fail completely.

The Cost of an Unseen Spectrum

Dropped Connections

Over **70%** call failures in affected areas.

Constant degradation of essential Mobile Money transactions for rural consumers.

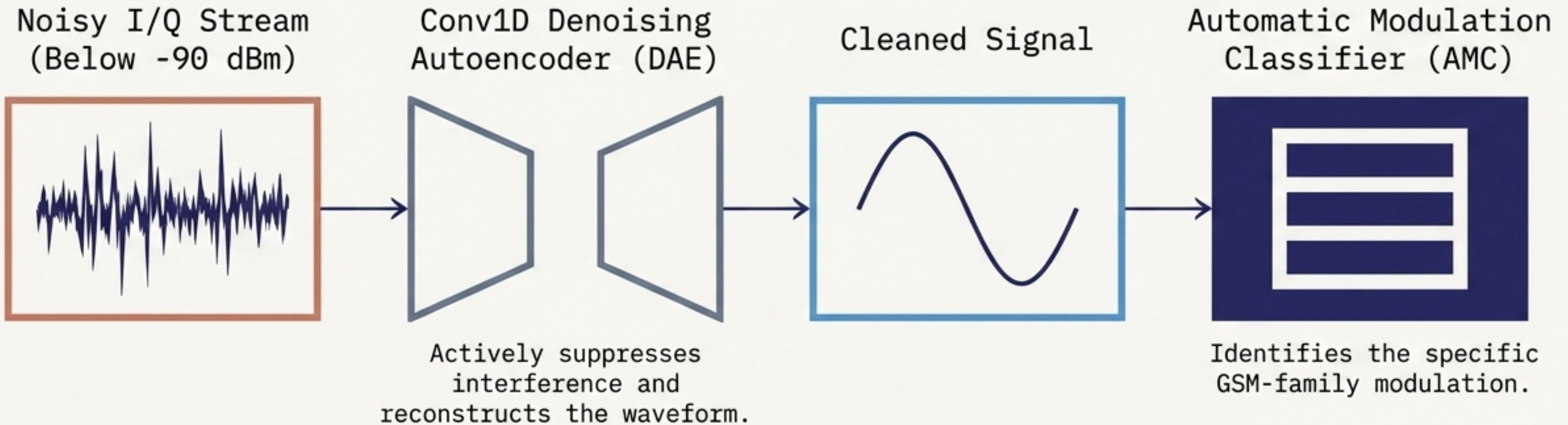
Regulatory Blind Spots

The Uganda Communications Commission holds a zero-tolerance policy, but current monitoring tech cannot “see” through the noise to locate illegal Bizindaalo or booster equipment.

Aviation & Safety Risks

Unmanaged spectrum interference continuously threatens critical aviation and public-safety communication channels.

The Solution: A Hybrid DAE-AMC Pipeline



Instead of forcing a classifier to read through noise, the system actively cleans the signal first—enabling accurate identification even in the most remote sub-counties.

Core Project Objectives

01



Design

Define the hybrid architecture, curate the RadioML/Kaggle datasets, and mathematically model the Uganda-specific interference conditions (boosters, Bizindaalo, rain fade).



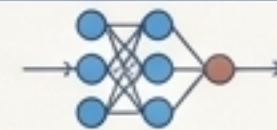
02

Implementation

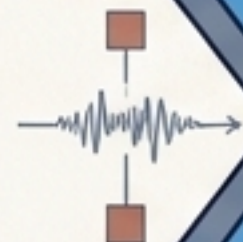
Develop and train the Conv1D DAE and the supervised AMC classifier for GSM modulations. Integrate both into a single Python/PyTorch pipeline.



Conv1D DAE



AMC Classifier



Python/PyTorch Pipeline



03

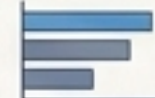
Analysis

Empirically evaluate the hybrid pipeline against a standalone AMC MC baseline using accuracy curves, F1-scores, and SNR-wise trends.

Accuracy Curves



F1-Scores



SNR-wise Trends

Accuracy Curves



The Evolution of AMC (Literature Synthesis)

The Brittle Past: Classical Methods

- Relied on **manual statistical analysis** and engineered features.
- Fatal Flaw:** Fails completely under **unknown channels** and **congested deployments** (as documented by UCC QoS audits).



The Modern Standard: Deep Learning

Models learn features directly from raw I/Q samples down to 0 dB.



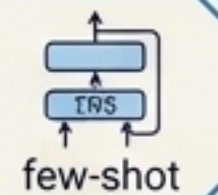
AbdElaziz et al. (2023): CNN with asymmetric kernels hits 96.5% at 0 dB.



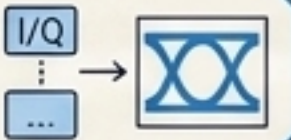
Zhang et al. (2023): MoE-AMC mixture-of-experts yields 71.8% accuracy.



Meta-Transformer (2024): Few-shot learning for unseen modulations.



Rehman et al. (2025): Converts I/Q streams to eye diagrams to overcome accuracy ceilings at -10 dB.



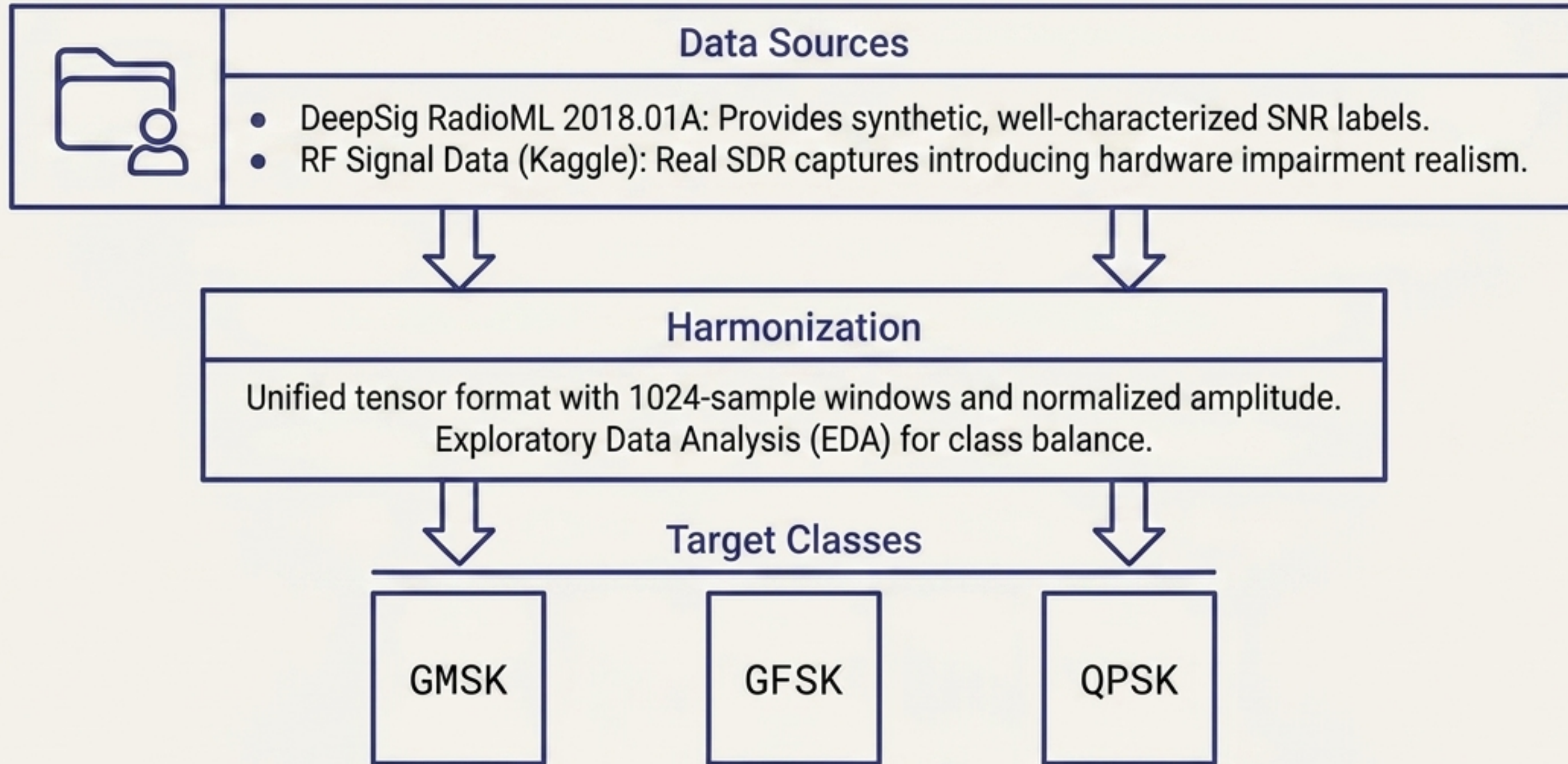
Defining the Research Gap



Capabilities	Classical Methods	Standard ML AMC	DAE-Enhanced AMC	Our Proposed Model
Works at -10 dBm SNR?			✓	✓
Handles Multi-modal interference?			✓	✓
Trained on Uganda-Specific Noise Vectors (Bizindaalo/Boosters)?				✓

DAEs are proven: Faysal et al. sustained 77.5% accuracy at -10 dB, and An & Lee delivered 70% relative gains. However, no published work applies this to African GSM networks' specific UCC-documented interference environment.

Methodology Phase 01: Design & Data



Focusing purely on the GSM-family modulations critical to rural basic phones.

Methodology Phase 02: Model Implementation



Conv1D Encoder-Decoder (The Cleaner)

Trained on noisy/clean I/Q pairs.
Optimized using MSE-based reconstruction loss to actively rebuild the signal waveform.

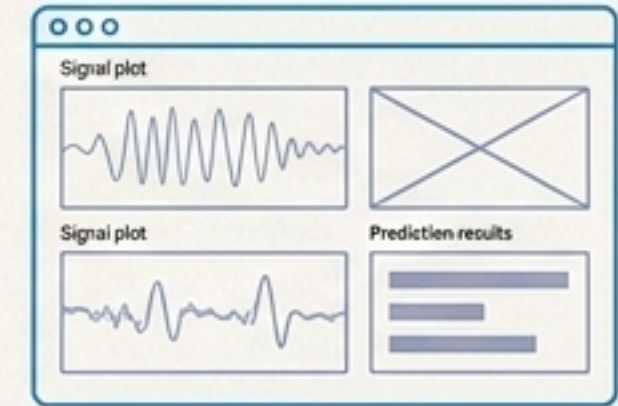
Architecture: Conv1D-based AE
Loss: MSE
Input: I/Q Pairs



Supervised AMC Classifier (The Reader)

Trained strictly on the output of the Denoising Autoencoder to recognize GSM-family signals (GMSK, GFSK, QPSK).

Input: Denoised IQ
Classes: GMSK, GFSK, QPSK
Loss: Categorical Cross-Entropy



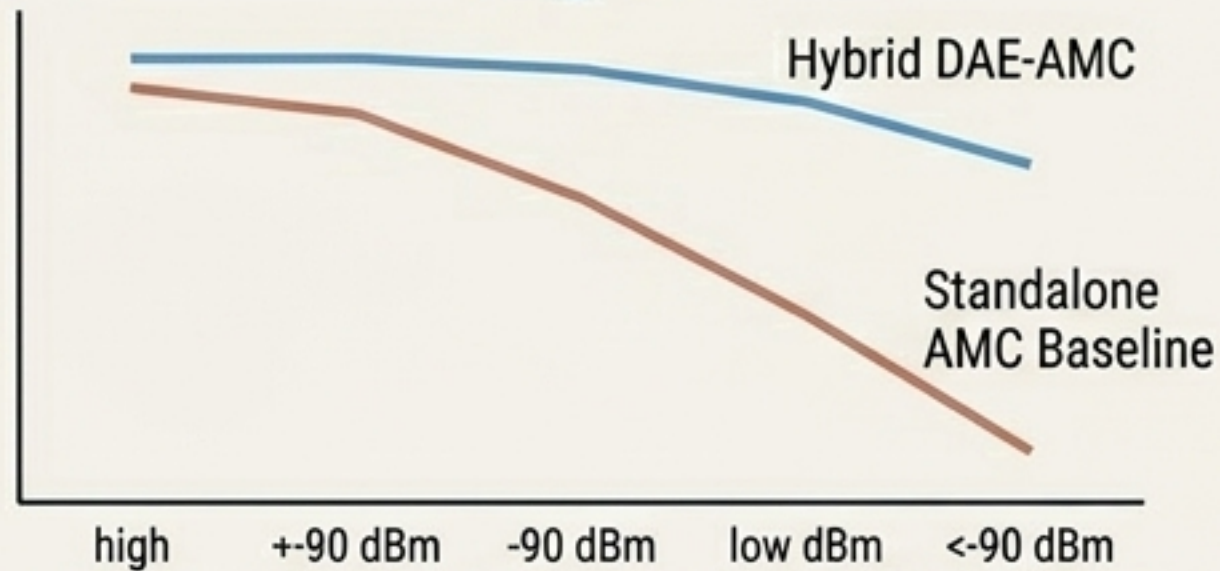
Integration & UI

Combining models into a hybrid pipeline (comparing frozen vs. fine-tuned weights).

Building a lightweight prototype interface to visualize noisy inputs, denoised outputs, and predictions.

Pipeline: Hybrid
Output: Visualization & Prediction
Interface: Prototype

Methodology Phase 03: Analysis & Benchmarking



Evaluation Criteria: Benchmark the hybrid DAE-AMC system directly against a standalone AMC baseline model across held-out SNR ranges (specifically testing the sub -90 dBm / low SNR conditions).



	GMSK	GFSK	QPSK
GMSK	70	10	13
GFSK	14	75	11
QPSK	15	10	100

Success Metrics: Accuracy Curves, Macro F1-Scores, Confusion Matrices, and SNR-wise performance trend comparisons.

Ultimate Goal: Translate statistical wins into rural monitoring capabilities (interference diagnosis for regulators).

Project Boundaries & Scope

Constraint = Environment

Offline Experimentation

Conducted entirely on compute hardware using pre-existing data for controlled testing.

Constraint = Stack

Python / PyTorch

Utilizing Jupyter Notebooks and standard cloud visualization libraries.

Constraint = Timeline

6 Months

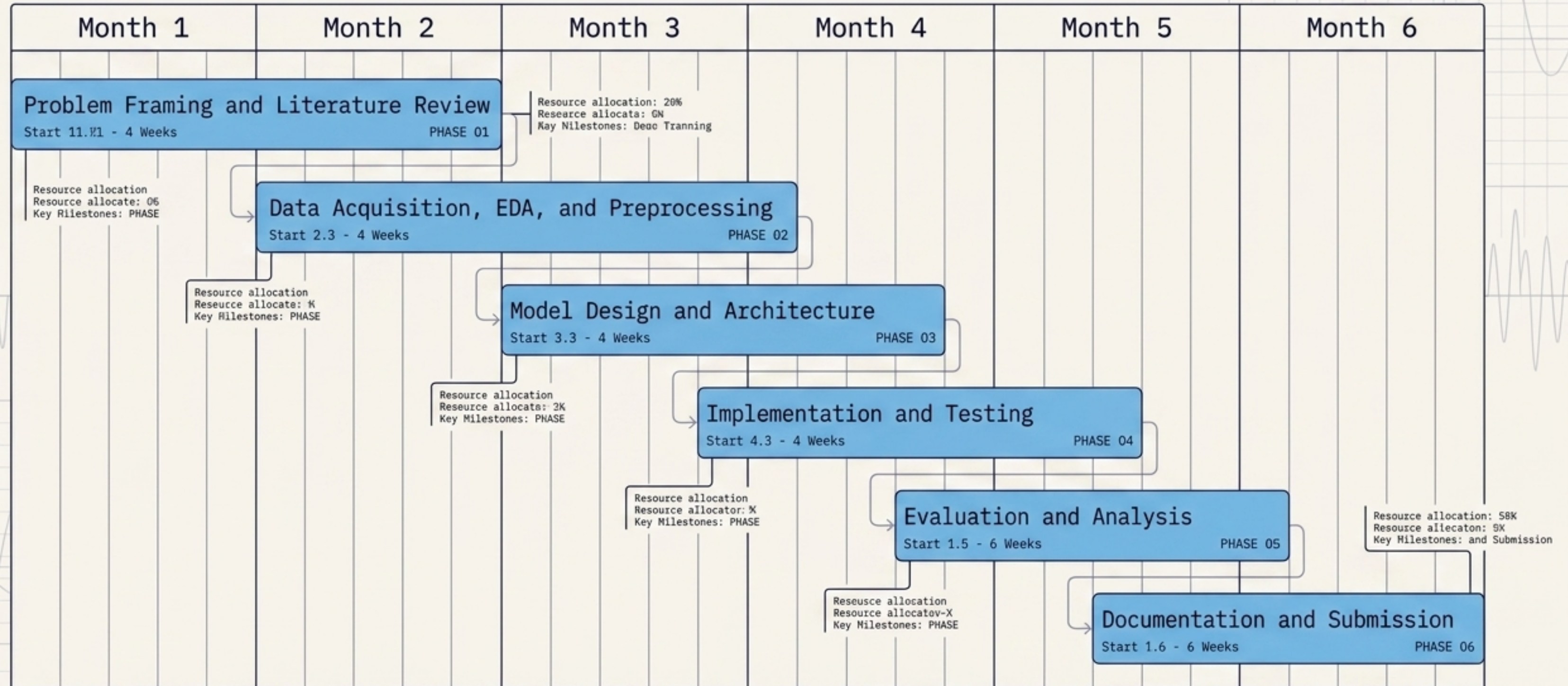
Dedicated, finite period for research, development, and testing phases.

Constraint = Targets

GSM-Family Only

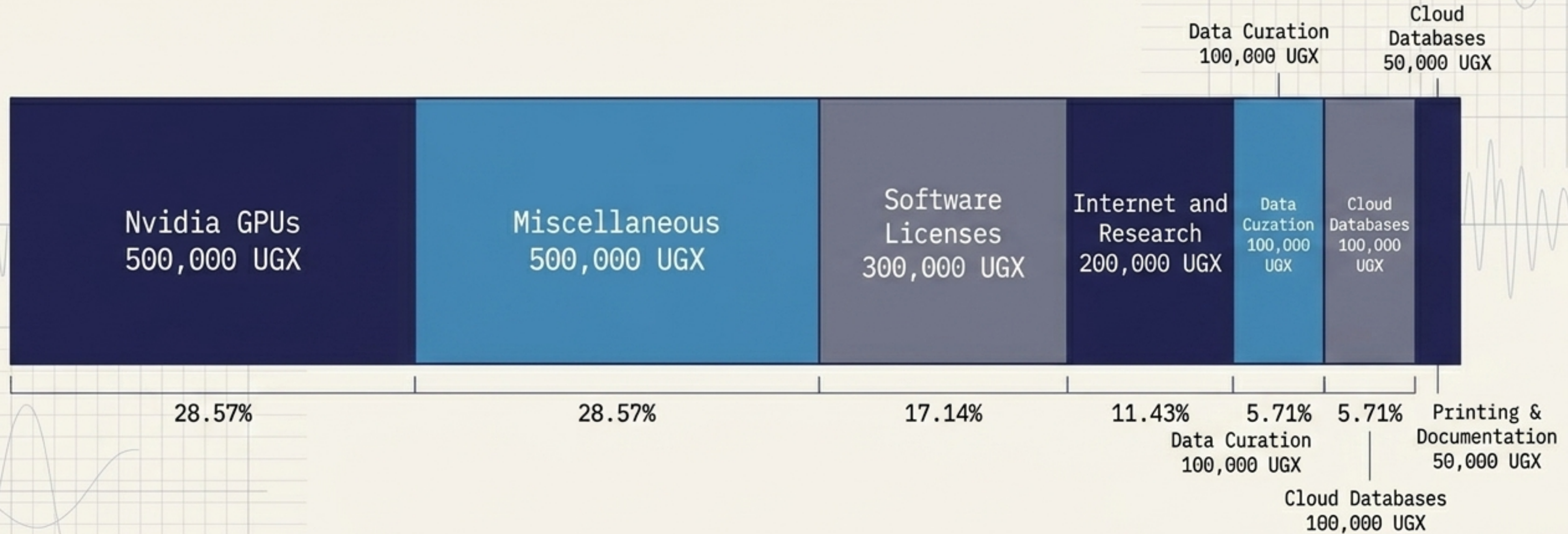
Specifically isolating legacy 2G protocols relevant to Ugandan basic phone users.

Execution Roadmap



Resource Allocation

Total: 1,750,000 UGX



Fully funded to support technical infrastructure, cloud data processing, and operational requirements.

Strategic Significance

Spectrum Surveillance



Empowers the UCC with software-based AI tools to finally detect illegal transmitters, confiscate boosters, and protect aviation channels.

Rural Broadband



30M
Lacking Access

Maintains reliable service in low-SNR environments, directly supporting the 30 million Ugandans currently lacking mobile internet access.

Economic Stability



Secures the radio links required for vital Mobile Money and e-government services in remote sub-counties.

Aligns fully with Uganda's **National Development Plan (NDP III)**, replacing brittle hand-crafted methods with robust, spectrum-efficient AI.

REF: NDPIII/AI-SPEC